

Lei Jiang

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EDUCATION

Ph.D. in Computer Science

University of Illinois Chicago, Chicago, IL

2024.9 — 2029.5(expected)

BA in Chemical Engineering

Sichuan University, Chengdu, Sichuan

2020.9 — 2024.6

RESEARCH EXPERIENCE

I'm generally interested in improving the generalization of large language models across various domains, including text generation and healthcare, by focusing on the representation within the models and also solving the generalization problem through fine-tuning. I'm good at using Python and related tools for machine learning and deep learning.

Natural Language Processing Laboratory

2024.12-present

Advised by [Dr.Parde](#), working on language model generalization and linear representation hypothesis.

R2 Lab

2024.9-2024.12

Advised by [Dr.Cheng](#), working on social bias and the uncertainty of large language models.

Interactive Image Segmentation

2023.1-2023.3

As an important part of lifelong learning automation closed Loop in KubeEdge SIG AI

- Survey: Investigate papers on interactive image Segmentation
- Proposal: Finish a proposal of integrating interactive image segmentation into KubeEdge SIG AI project

PUBLICATION

What Do LLMs Know About Alzheimer's Disease? Fine-Tuning, Probing, and Data Synthesis for AD Detection.

Lei Jiang, Yue Zhou, Natalie Parde. Submitted to ACL 2026(under review).

Threshold Filtering Packing for Supervised Fine-Tuning: Training Related Samples within Packs. Jiancheng Dong,

Lei Jiang, Wei Jin, Lu Cheng. In NAACL'25 Main Conference

Diagnosing Moral Reasoning Acquisition in Language Models: Pragmatics and Generalization. Guangliang Liu, Zimo Qi, Xitong Zhang, **Lei Jiang**, Kristen Marie Johnson. In EMNLP'25 Findings.

PROJECTS

SGLang

2026.1-present

SGLang is a high-performance serving framework for large language models and multimodal models. I contributed on:

- Developing inference engine testing.
- Systematic profiling of multi-turn training with Verl and observing the long-tail effect in rollout.

KubeEdge-Sedna Python SDK Integration with MindSpore

2022.8-2022.10

Sedna is an **open-source** edge-cloud synergy AI project running on K8S, supporting functions like incremental learning and federated learning, but doesn't support users to facilitate its edge-cloud synergy capabilities with the model developed by MindSpore deep learning framework

- Integrate MindSpore deep learning framework in Sedna lib.
- Implement [a classification demo based on the Mindspore framework](#), with incremental learning ability on the K8S cluster.
- Write [the proposal of integrating TinyMS framework](#), which runs on top of MindSpore and provide high-level API

Kaggle Leaves Classification

2022.7

The purpose of this competition is to classify 176 categories of leaves and achieve as high as possible accuracy. Here is what I do:

- Implement the [whole framework](#) for preprocessing, training, and inference from scratch with PyTorch.
- Test a variety of models' performance considering the different sizes and structures and choose SEResNet.
- Use techniques like data augmentation, cross-validation, and multimodel inference to achieve 0.98159 accuracy.

Edge Computing Based Removable Surveillance Alarm Camera

2021.10-2022.7

Considering the problem that in temporary construction sites it's expensive and time-consuming to set up a complete surveillance camera system, I propose to use edge computing technique to implement a plug-and-play camera with alarm function. I finish:

- Implement image preprocessing with OpenCV python lib.
- Use yolov5 to recognize a series of wearable objects like masks, safety helmets, protective clothing
- Transfer it to Jeston Nano as the computing equipment, using the USB camera as the input.

HONOR & AWARD

- ALCF Director's Discretionary (DD) allocation award from Argonne Leadership Computing Facility
- 2022 Open Source Promotion Plan Most Promising Award (1/5 in 449 students)
- Huawei-ZhiNengJiZuo Fellowship 2022 (1/30 in the whole university)